The Acceleration of Climate Risk: A Fifty-Year Longitudinal Analysis of Extreme Weather Socio-Economic Impacts (1975–2024)

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Abstract. This study investigates the evolution of extreme weather event frequency and their socio-economic impacts over a fifty-year period (1975–2024). Utilizing a consolidated dataset from EM-DAT, NOAA, and Our World in Data, we apply exploratory data analysis and interactive visualization to identify structural shifts across two distinct 25-year blocks. Our findings reveal a significant "densification" of high-severity disasters and a transition from rare economic shocks to persistent financial burdens, particularly in the 0.3% to 0.8% GDP loss range. The results identify North America as the region with the steepest economic acceleration (340.28%), while Asia remains the global epicenter of human impact with 3.6 billion people affected in the 2000–2024 period. Despite a moderate correlation (r = 0.33) between temperature anomalies and GDP losses, a critical surge in mortality was observed in Europe, highlighting that even high-income regions face increasing lethal risks. This research provides empirical evidence of a shifting risk landscape, emphasizing the urgent need for regionalized adaptation strategies as climate impacts transition from isolated events to a systemic and accelerating drag on continental growth.

Keywords: Climate Change · Extreme Weather Events · Data Visualization · Socio-Economic Impact · Regional Acceleration · GDP Loss · EM-DAT · Interactive Dashboards.

1 Introduction

Anthropogenic climate change has emerged as one of the most critical challenges of the 21st century, fundamentally altering the frequency and intensity of extreme weather events across the globe [1]. The accumulation of greenhouse gases (GHG) in the atmosphere has led to significant positive temperature anomalies, which act as catalysts for hydrometeorological and climatological disasters such as floods, storms, and wildfires. While the physical mechanisms of these events are well-documented, their socio-economic repercussions—measured in terms of human mortality and Gross Domestic Product (GDP) losses—reveal a complex landscape of global vulnerability and varying regional resilience [2].

Recent studies indicate that the impacts of extreme weather are not distributed uniformly, often disproportionately affecting regions with limited infrastructure or high exposure to environmental hazards. Furthermore, the evolution of reporting systems over the last five decades has improved data availability, yet it also highlights a concerning trend: the "normalization" of high-impact events that were previously considered rare [3]. Understanding whether these patterns represent a steady increase or a sharp acceleration is vital for policy-making and disaster risk management.

This study leverages 50 years of historical data (1975–2024) to analyze the nexus between climate drivers and disaster outcomes. By comparing two distinct 25-year periods—1975–1999 and 2000–2024—this research aims to identify structural shifts in disaster patterns across six major continental regions. Specifically, this paper addresses the following research question: **How have the frequency and economic damage of extreme weather events changed over the past 50 years, and which world regions show the steepest acceleration?**

2 Related Work

The scientific consensus on the link between anthropogenic climate change and the intensification of extreme weather events is well-established in the literature. Foundational work by the Intergovernmental Panel on Climate Change (IPCC) provides a comprehensive framework for understanding how rising global mean temperatures influence the thermodynamic conditions of the atmosphere, leading to more frequent and severe hydrometeorological disasters. These reports emphasize that while natural variability exists, the signal of human influence is increasingly detectable in heatwaves, heavy precipitation, and drought patterns.

Recent studies [1] have advanced the field of event attribution, demonstrating that many individual extreme events would have been virtually impossible or significantly less intense without human-induced warming. Their research highlights that the impacts are not merely physical but are compounded by social vulnerabilities, where the severity of a disaster is often dictated as much by local exposure and adaptive capacity as by the meteorological event itself. This perspective is crucial for shifting the focus from purely physical climate metrics to human-centric impact assessments.

The economic dimension of these disasters is a primary concern for international policy. Ferreira [2] explores the long-term socio-economic consequences of extreme weather, arguing that significant capital destruction and labor productivity losses can hinder regional development for years following an event. Her work underscores a critical challenge in disaster research: the "reporting bias." Historical datasets, such as the EM-DAT database, often show a lower frequency of events in the late 20th century, which may be attributed to less sophisticated monitoring and communication infrastructure in developing regions rather than a true absence of events.

Despite these advancements, a gap remains in how multi-decadal regional acceleration is communicated to non-technical stakeholders. Much of the existing literature is presented through static reports and complex statistical models that may be inaccessible to policy-makers. This project seeks to fill this gap by applying exploratory data analysis (EDA) and interactive visualization techniques to a 50-year longitudinal dataset. By explicitly comparing 25-year blocks (1975–1999 vs. 2000–2024), this work provides a clearer visual narrative of how regional risks have evolved and accelerated, leveraging tools like Streamlit to democratize access to climate impact data.

3 Data & Methods

3.1 Dataset Description

This study utilizes a multi-source longitudinal dataset spanning from 1975 to 2024. The data was consolidated from three primary scientific repositories:

- **EM-DAT International Disaster Database:** Used to extract records of natural disasters, specifically focusing on droughts, floods, storms, extreme temperatures, and wildfires. The variables include absolute numbers of deaths and affected populations. Furthermore, **Economic Impact Data** was derived from EM-DAT's reported financial losses and subsequently standardized as a percentage of regional Gross Domestic Product (GDP) to allow for consistent cross-continental comparisons.
- **NOAA Global Surface Temperature Dataset:** Provided monthly and yearly regional temperature anomalies and surface temperature averages in degrees Celsius.
- **Our World in Data:** Sourced for greenhouse gas emissions (measured in tonnes per capita of CO_2 equivalent) and cleaned CSV versions of natural disaster impacts.

3.2 Data Pre-processing and Integration

The raw datasets were integrated into a unified master repository using Python's Pandas library. The merging process was performed using a two-key join operation based on "Year" and "Region". To address the research question regarding acceleration, the 50-year timeframe was segmented into two distinct 25-year blocks: 1975–1999 and 2000–2024. Geographic data was aggregated into six continental regions: Africa, Asia, Europe, North America, South America, and Oceania. Feature engineering was applied to calculate "Acceleration" metrics, defined as the percentage change in impacts between the two defined periods.

3.3 Visualization Approach and Justification

The visualization strategy follows Kirk's 5-step design process and the data-to-viz framework to ensure clarity for a non-technical audience.

- **Faceted Histograms:** Chosen to compare the distribution of mortality and frequency between the two periods. Faceting (small multiples) was preferred over overlapping bars to avoid visual clutter and clearly show shifts in the "long tail" of extreme events.
- **Correlation Matrix (Heatmap):** Employed during the exploratory phase to identify the strength of the relationship between GHG emissions, temperature anomalies, and socio-economic damages.
- **Scatter Plots with OLS Regression:** Used to visualize the causal-effect hypothesis between climate drivers and economic loss. The inclusion of trend lines and color-coding by region facilitates the identification of regional outliers and slopes of acceleration.
- **Box Plots:** Utilized for outlier detection and to visualize the dispersion and variability of GDP losses within each 25-year block.
- **Choropleth Maps:** Selected as the primary tool for geographic communication, providing an immediate visual identification of "hotspots" where climate-driven economic damage is most severe in the 2000–2024 period.

3.4 Reproducibility and Tools

The entire analysis is implemented in a Python environment utilizing the Plotly Express library for interactive graphics and Streamlit for dashboard deployment. All code, processed datasets, and the dependency environment (`requirements.txt`) are hosted in a public GitHub repository to ensure full scientific reproducibility.

4 Results

The analysis of the consolidated dataset (1975–2024) reveals a complex and sometimes counterintuitive relationship between climate variables and socio-economic impacts. The results highlight shifts in disaster patterns and the increasing financial burden on regional economies.

4.1 Correlation Between Climate Drivers and Impacts

The correlation matrix (see Fig. 1) provides a quantitative overview of the interdependencies between climate drivers and their consequences. Contrary to global aggregate expectations, the correlation between per capita greenhouse gas emissions and regional temperature anomalies was found to be near zero and slightly negative ($r = -0.06$) within this specific dataset. This suggests that, at a per capita and regional level, other factors or temporal lags may decouple local emission rates from immediate temperature deviations.

However, a moderate positive correlation ($r = 0.33$) was observed between **Yearly anomaly average** and **Total GDP Loss**. This indicates that as temperatures deviate further from the historical baseline, the economic impact of disasters tends to increase proportionally. In contrast, mortality shows a near-zero correlation with temperature anomalies ($r = 0.01$), suggesting that the number of deaths is likely driven by specific catastrophic events or varying levels of

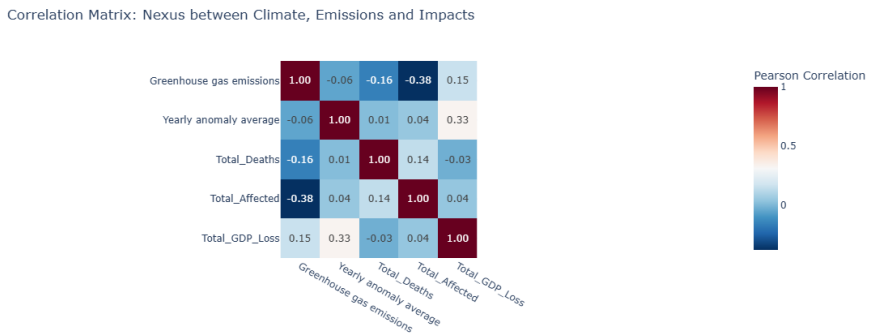

Fig. 1. Correlation Matrix: Nexus between Climate, Emissions and Impacts

regional preparedness rather than gradual temperature increases. Additionally, a negative correlation between emissions and affected populations (r = -0.38) indicates that regions with higher per capita emissions in this dataset (often more developed) tend to report fewer people affected, potentially due to better infrastructure and resilience.

4.2 Shifts in Mortality Patterns

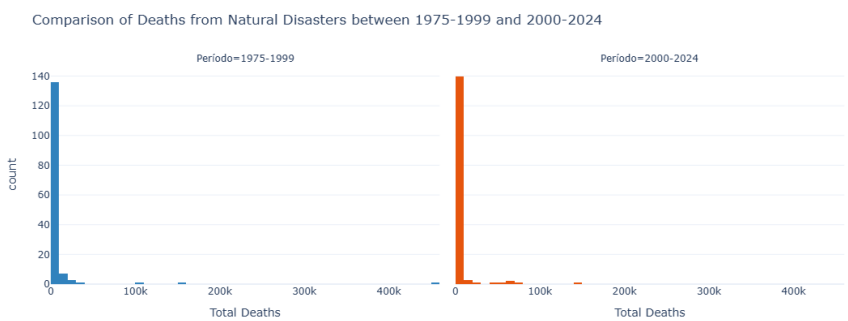

Fig. 2. Comparison of Deaths from Natural Disasters between 1975-1999 and 2000-2024

A comparative analysis of mortality distributions between 1975–1999 and 2000–2024 (see Fig. 2) reveals a shift in disaster severity. Although the historical peak of mortality occurred in the first period (exceeding 450,000 deaths in a single event cluster), the 21st century is characterized by a "densification" of high-lethality events.

Specifically, the period of 2000–2024 recorded a higher frequency of events in the intermediate-high range. While the 1975–1999 block had only three events in the 20k–30k death range and jumps to extreme outliers, the recent period exhibits a more continuous distribution of severe disasters, with six events clustered between 40,000 and 80,000 deaths. This suggests that while absolute historical extremes have not been surpassed, the probability of encountering a "major" disaster has increased.

4.3 Escalation of Economic Vulnerability

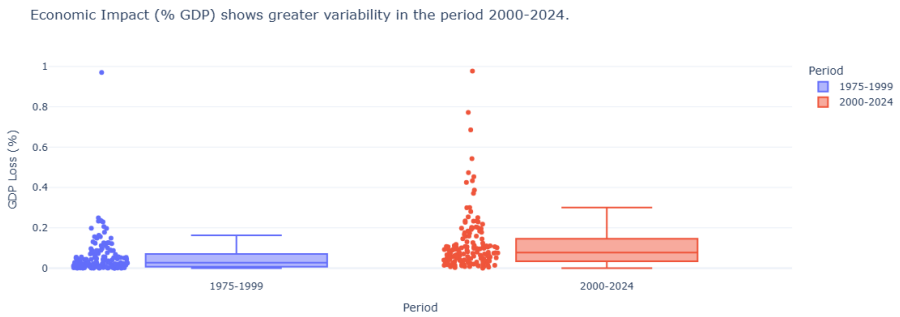

Fig. 3. Economic Impact (% GDP) shows greater variability in the period 2000-2024.

The economic impact, measured as a percentage of regional GDP, provides the clearest evidence of climate acceleration. As shown in the box plots comparing both 25-year blocks (Fig. 3), there is a notable "spreading" of economic damages in the 2000–2024 period.

The data indicates that the number of years with "zero registered impact" decreased from four in the first period to only one in the second. Moreover, while both periods had extreme events nearing 1% of the regional GDP, the second period shows a much higher frequency of losses in the 0.3% to 0.8% range. This indicates that climate-related costs are moving from being "rare catastrophic shocks" to becoming a "persistent economic drag" on continental growth.

4.4 Regional Acceleration Hotspots

The geographic distribution of impacts reveals a highly heterogeneous landscape of climate acceleration. When calculating the percentage change in GDP loss between the two periods, North America emerged as the region with the steepest economic acceleration, recording a staggering 340.28% increase, from 1.40% of regional GDP in 1975–1999 to 6.18% in 2000–2024 (see Fig. 4). Africa and Asia

Region	Acceleration (% GDP)
North America	340.277408
Africa	94.883048
Asia	92.867151
Oceania	81.936203
Europe	47.893920
South America	34.245024

Table 1. Acceleration of Economic Impact by Regions (%)

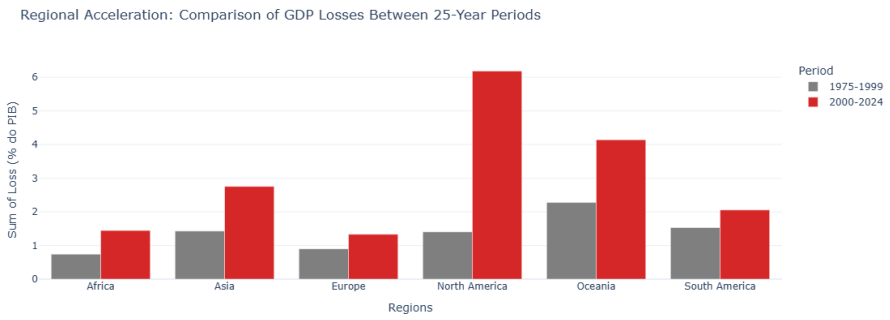

Fig. 4. Regional Acceleration: Comparison of GDP Losses Between 25-Year Periods.

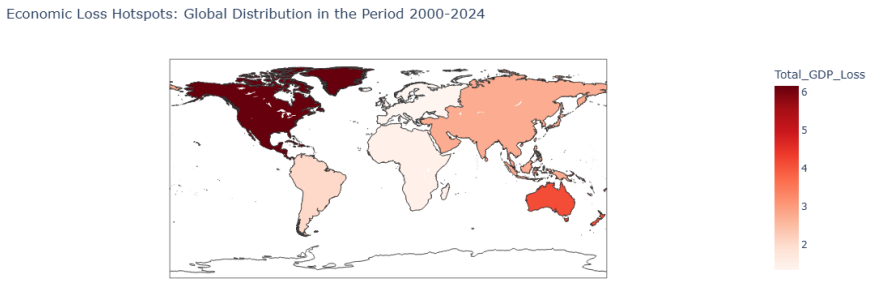

Fig. 5. Economic Loss Hotspots: Global Distribution in the Period 2000-2024.

also demonstrated significant escalations in economic burden, with losses nearly doubling (94.88% and 92.87% acceleration, respectively), like show Table 1.

In terms of the human toll, Asia remains the global epicenter of climate vulnerability, with over 3.6 billion people affected in the 2000–2024 period, despite a slight decrease in total mortality compared to the previous block. Conversely, Europe presented a critical anomaly: while its economic acceleration was more moderate (47.89%), the region experienced a dramatic surge in total deaths, jumping from approximately 6,500 in the first period to over 328,000 in the second. This suggests that while economic resilience might be higher in developed regions, specific event types—likely extreme temperature waves—are causing unprecedented lethality. Overall, these findings (visualized in Fig. 5) confirm that while the climate crisis is global, the acceleration of its socio-economic consequences is disproportionately concentrated in North America, Africa, and Asia.

5 Discussion

The results of this study confirm a significant acceleration in the socio-economic impacts of extreme weather events over the last 50 years. The most striking finding is the decoupling of economic damage from human mortality in certain regions. For instance, the **340% increase in GDP loss in North America** contrasts with a decrease in reported deaths, suggesting that while infrastructure and economic assets are increasingly exposed to climate risk, improved emergency response and early warning systems may be mitigating mass-casualty events.

However, the "densification" of events in the 40k–80k mortality range during the 2000–2024 period indicates that the global system is facing a new tier of "high-frequency, high-impact" disasters. The moderate correlation ($r = 0.33$) between temperature anomalies and GDP loss provides empirical evidence that regional warming is a persistent driver of economic instability. In Europe, the staggering increase in mortality serves as a warning that even high-income regions are not immune to the lethal effects of climate-driven extreme temperatures.

6 Limitations and Ethics

The interpretation of multi-decadal longitudinal data at the intersection of climatology and socio-economics requires a critical evaluation of the underlying datasets and the environmental contexts in which they were recorded. While the findings presented in this study provide a compelling narrative of accelerating climate risk, the inherent complexities of disaster reporting and the evolution of global monitoring systems necessitate a transparent disclosure of methodological constraints. Ensuring scientific integrity involves acknowledging that the observed trends are shaped not only by physical atmospheric shifts but also by the socio-technical progress in how humanity detects, records, and values loss.

Consequently, the following limitations must be considered when generalizing these results:

- **Reporting and Documentation Bias:** The apparent surge in disaster frequency during the 21st century may be partially influenced by the "digital revolution" in data collection. Modern satellite monitoring and real-time reporting infrastructure are far superior to the fragmented systems available between 1975 and 1990, potentially leading to an artificial inflation of recent records compared to historical baselines.
- **Scale and Aggregation Fallacy:** Analyzing data at a continental level inevitably conceals severe intra-country inequalities. A single high-magnitude event in one nation can skew the statistics for an entire continent, potentially masking successful local adaptation strategies or localized crises in neighboring countries.
- **Economic Value Normalization:** Although loss is measured as a percentage of GDP, the absolute density of capital and infrastructure in high-risk zones (such as coastal cities) has increased exponentially since 1975. This "wealth concentration" effect means that even if the physical intensity of a storm remains constant, the resulting economic damage is naturally higher today than in previous decades.
- **The Ecological Fallacy:** It is crucial to note that the correlations identified at the continental level do not imply a direct causal link for individual events. The relationship between temperature anomalies and mortality, for instance, is mediated by complex variables such as public health infrastructure, urban planning, and emergency response capabilities which were not fully captured in this quantitative model.

7 Conclusion

This research applied data visualization and exploratory analysis to investigate the evolution of extreme weather impacts. Our findings show that the world has transitioned from a period of rare, extreme shocks to one of persistent and accelerating climate-driven costs. North America shows the fastest economic acceleration, while Asia continues to suffer the highest human impact. These results underscore the urgent need for regionalized adaptation strategies that account for the specific vulnerability profiles identified in this study.

Acknowledgements The authors would like to express their sincere gratitude to Professor Maryam Abbasi for her initiative and for proposing the research topic, which was fundamental to the development of this study. We also thank her for the valuable knowledge and guidance shared throughout the process. Furthermore, we would like to acknowledge the support provided by the Santarém Polytechnic University.

References

1. Clarke, B., Otto, F., Stuart-Smith, R., Harrington, L.: Extreme weather impacts of climate change: an attribution perspective. Environmental Research: Climate **1**(1), 012001 (jun 2022). https://doi.org/110.1088/2752-5295/ac6e7d
2. Ferreira, S.: Extreme weather events and climate change: Economic impacts and adaptation policies. Annual Review of Resource Economics **16**(Volume 16, 2024), 207–231 (2024). https://doi.org/https://doi.org/10.1146/annurev-resource-101623-095314
3. IPCC: Sections. In: Lee, H., Romero, J. (eds.) Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, pp. 35–115. IPCC, Geneva, Switzerland (2023). https://doi.org/10.59327/IPCC/AR6-9789291691647